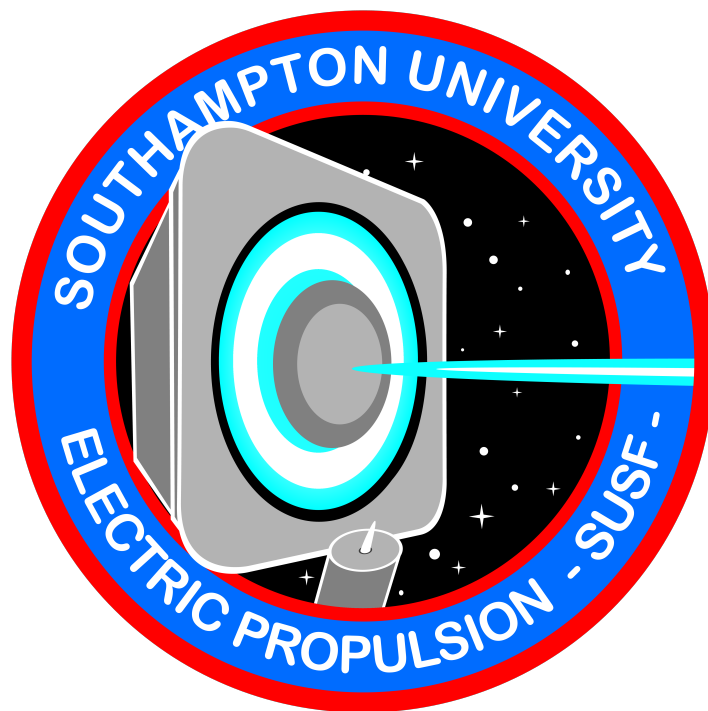


Io-2 Documentation



Southampton University Electric Propulsion

March 2026

1 Introduction

Io-2 is the natural successor to Io-1, building on its legacy. Io-1 was first successfully ignited in February 2025. Io-1 could not operate steadily, and it was not possible to acquire hard data points. Despite this, many certain aspects of its performance can be inferred.

Based on the magnetic field tests before Io-1's test fire, the magnetic field appeared lopsided. This resulted in a non-ideal field topology for optimum performance. An ideal field should be mostly radial. This resulted in the plume being quite divergent, as shown in the test fire images in Figure 1. This likely caused many ions to have significant non-axial components in their velocities, in turn reducing the overall thrust.

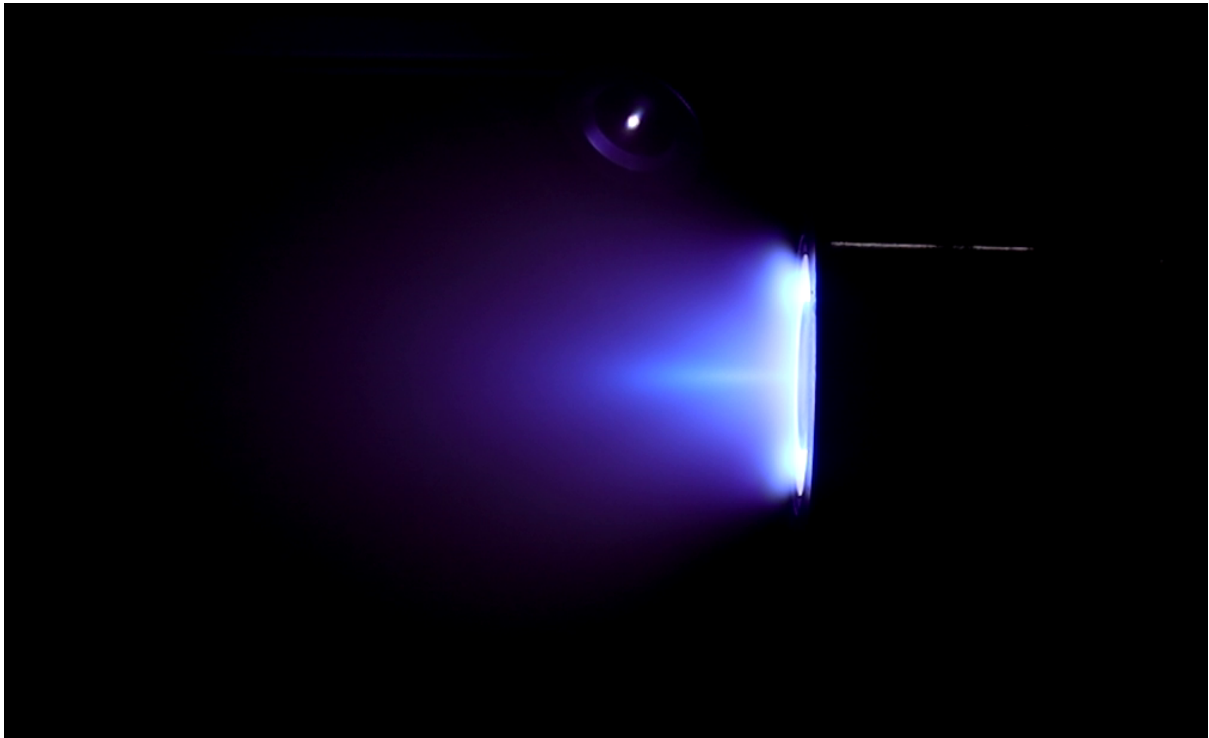


Figure 1: Io-1 operating on Krypton gas at the David Fearn Electric Propulsion Laboratory.

Another observation is that the plasma appears very dense near the exit plane of the thruster. This indicates that most of the ionisation and acceleration is occurring close to the exit plane, explaining the instabilities and poor performance. As a result of this, for Io-2, great care was taken to ensure a highly radial field with magnetic lensing to ensure the greatest performance by keeping the magnetic field strength lower in the channel. Additionally, an effort has been made at attempting to magnetically shield Io-2 to reduce wall erosion, thermals and improve overall efficiency.

Table 1: Input design parameters for Io-2.

Parameter	Value
Nominal Discharge Power	1000W
Nominal Discharge Voltage	300V
Propellant	Argon
Project Budget	£1600

2 Channel Sizing

The discharge channel dimensions of a Hall-Effect thruster (HET) are one of the most important design parameters which effects a HET's performance. The channel width and diameter are the two critical features we often pay close attention to in the design phase. The plasma has strong interactive effects with the discharge channel walls, and so changing the channel width and mean channel diameter often has strong performance effects. It has been found from the literature [1] that increasing the channel width to mean channel diameter tends to give a significant performance increase.

To size the cross-sectional discharge channel of a Hall-effect thruster, it is common practice to utilise scaling laws. Often, a database containing geometric parameters along with performance metrics is used. Due to the small scale of our team, we used Lee's database [2]. The database assumes all thrusters are operating on xenon and that in the centre of the discharge channel there is a peak magnetic field density of $200G$. The scaling laws derived can be used to define the mean channel diameter and channel width based on an optimum design power and discharge voltage. Lee's power scaling equation lets the discharge power become just a function of mean channel diameter d and discharge voltage V_d ,

$$P = C_P d^2 V_d. \quad (1)$$

Where C_P is a power scaling coefficient used to correlate the variables. This equation is used to determine the mean channel diameter of the thruster. To determine the channel width, a second scaling equation can be used which correlates the mean channel diameter directly to the channel width h .

$$h = C_{hd} d. \quad (2)$$

Where C_{hd} is the channel width to mean channel diameter scaling coefficient. The mean channel diameter is defined by:

$$d = (d_o + d_i)/2 \quad (3)$$

where d_i is the inside diameter, d_o is the outside diameter.

Unlike for the discharge channel cross section, there is no reliable scaling relation for the channel length. Therefore, a set of constraints is used to provide a guideline for its length. The constraints exist as:

$$\lambda_i = \frac{v_n}{n_n \langle \sigma_i \nu_e \rangle} \ll L \quad (4)$$

Where λ_i is the ionisation mean free path in mm, v_n is the neutral particle's thermal velocity, n_n is the neutral number density, $\langle \sigma_i \nu_e \rangle$ is the ionisation reaction rate, and L is the discharge channel length. Neutral particles must have a significant distance to travel across to become ionised, to increase the probability of this occurring, the constraint given by Equation 4 exists.

$$r_{L,e} = \frac{m_e v_{th}}{|q|B} \ll L \quad (5)$$

Where $r_{L,e}$ is the electron Larmor radius, m_e is the mass of an electron, v_{th} is the mean thermal speed of the electrons, q is the charge, and B is the peak magnetic flux density. Electrons must be significantly confined to disrupt their motion towards the anode, increasing the probability of collisions with neutrals to form ions. The mean thermal velocity of the electrons is expressed by averaging the Maxwell-Boltzmann distribution, giving

$$v_{th} = \sqrt{\frac{8k_B T_e}{\pi m_e}}. \quad (6)$$

Where k_B is the Boltzmann constant, and T_e is the electron temperature in kelvin. Similarly, the ion Larmor radius is

$$r_{L,i} = \frac{m_i v_i}{|q|B} \gg L, \quad (7)$$

Where m_i is the ion mass, and v_i is the ion velocity, which can be assumed to be approximately the same as the exhaust velocity,

$$v_i = v_e = \sqrt{\frac{2qV_d}{m_i}}. \quad (8)$$

Unlike electrons, ions should be weakly magnetised to maintain a mostly axial trajectory from the discharge channel.

Using work done by Thomas [3], the reaction rate coefficient can be predicted for a given element across a range of electron temperatures.

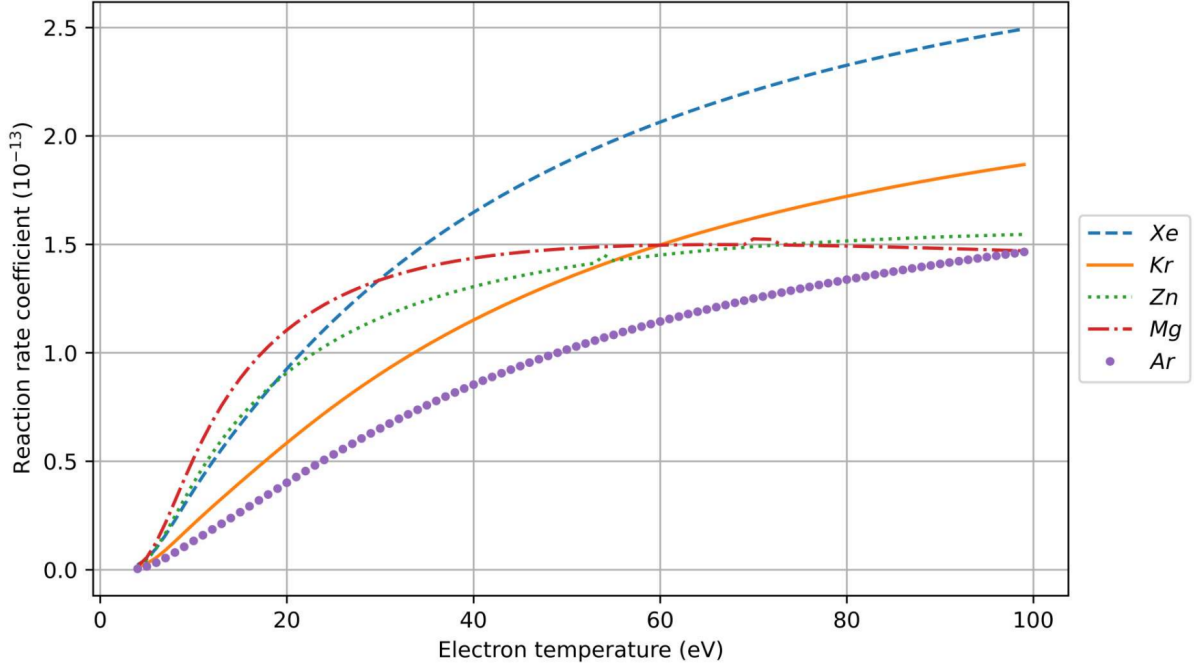


Figure 2: The calculated reaction rate coefficient against electron temperature for propellants: Xenon, Krypton, Zinc, Magnesium and Argon [3]

$$T_{eV} = 0.12V_d \quad (9)$$

Equation 9 is an empirical relationship to determine the bulk electron temperature from the discharge voltage. Figure 2 along with Equation 9 allows an estimate of the reaction rate coefficient to be calculated. The ionisation mean free path for argon can be calculated.

Along with modelling the discharge channel, Lee's database allows for high-level performance prediction by correlating the thrust and mass flow rate to certain parameters. Mass flow rate is directly related to the channel cross section,

$$\dot{m} = C_{\dot{m}} h d. \quad (10)$$

Where $\dot{m}$ is the anode mass flow rate, and $C_{\dot{m}}$ is the associated scaling coefficient. Thrust is then related to the mass flow rate and discharge voltage,

$$T = C_T \dot{m} \sqrt{V_d}. \quad (11)$$

Where C_T is the thrust scaling coefficient.

When considering operation on Argon, the relationship between thrust and specific impulse needs to be scaled. The Argon performance is predicted by relating the exhaust velocity and thrust equations,

$$\frac{T_{Ar}}{T_{Xe}} = \frac{I_d \sqrt{\frac{2qV_d}{m_{Ar}}}}{I_d \sqrt{\frac{2qV_d}{m_{Xe}}}} = \sqrt{\frac{m_{Xe}}{m_{Ar}}}. \quad (12)$$

Similarly, to thrust, the specific impulse for an Argon plasma discharge can be found from,

$$\frac{I_{sp,Ar}}{I_{sp,Xe}} = \sqrt{\frac{m_{Ar}}{m_{Xe}}} \quad (13)$$

Using the scaling coefficients and reasoning above, the following channel parameters and expected performance metrics are predicted for Io-2 when operating on Xenon and Argon.

Table 2: Lee scaling coefficient values

Coefficient	Value	Units
C_P	633.0	$\frac{W}{Vm^2}$
C_{hd}	0.242	n/a
$C_{\dot{m}}$	0.003	$\frac{kg}{m^2s}$
C_T	892.7	$\frac{NsV^{0.5}}{kg}$

Table 3: Input and output parameters for the discharge channel and expected performance

Parameter	Value	Unit
Discharge Power	1000	W
Discharge Voltage	300	V
Peak Magnetic Flux Density	200	G
Bulk Electron Temperature	36	eV
Ionisation Mean Free Path	0.479	mm
Electron Larmor Radius	1.140	mm
Ion Larmor Radius	249.499	mm
Mean Channel Diameter	72.6	mm
Channel Width	17.6	mm
Channel Length	35.0	mm
Discharge Current	3.33	A
Anode Mass Flow Rate	3.833	mg/s
Thrust (Xenon)	59.3	mN
ISP (Xenon)	1576.2	s
Thrust (Argon)	32.7	mN
ISP (Argon)	2857.4	s

3 Magnetic Field Design

The magnetic field topology has multiple operational functions within the HET. The magnetic field serves to reduce the mobility of the electrons, effectively slowing them down. This gives more opportunity for ionisation of the neutral propellant to occur. The second operational feature is that it acts as the force transfer mechanism. Ions are accelerated out of the channel by the electric field. However, it is through

the Hall effect, created by the $E \times B$ of the fields, that the accelerating force exerts an equal and opposite force back onto the thruster.

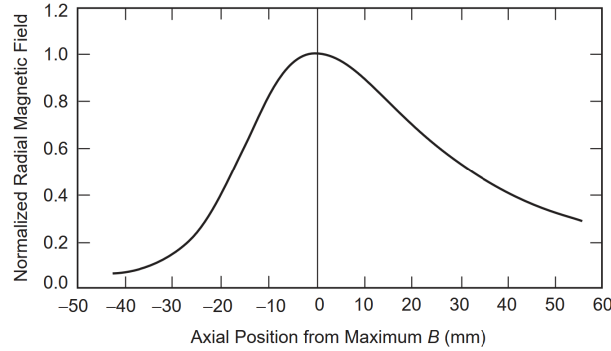


Figure 3: Axial variation in radial magnetic field normalised to the peak radial field in the NASA-173Mv Hall thruster [4].

Traditional Hall effect thrusters have a topology such that the radial field component slowly rises from zero at the anode, rapidly increases in strength until peaking at the channel exit, or just before. The field strength then traditionally tapers off. This can be visualised in Figure 3.

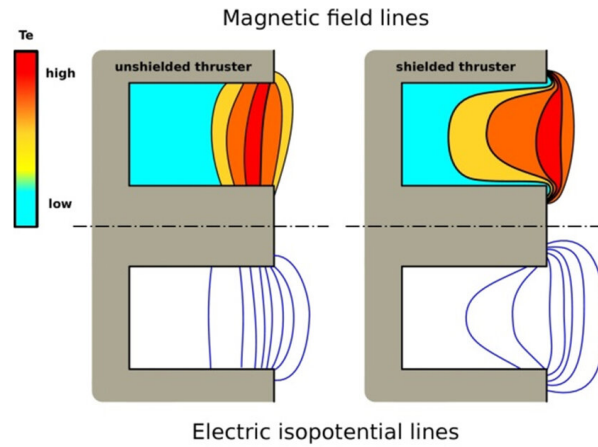


Figure 4: Illustration of the magnetic field isolines along with the electron temperature distribution in the channel for unshielded topologies (left) and shielded (right)[5] .

Thrusters are either magnetically shielded (MS) or unshielded (US). Magnetic shielding was developed as a method to extend the lifetimes of Hall-effect thrusters by reducing the ion energy before it impacts the channel walls. Reduced ion energy results in reduced channel wall erosion rates. Io-2's topology was designed with the intention of being magnetically shielded. A 2D cross-section of the thruster was iterated using FEMM to simulate the magnetic field for different configurations of the magnetic circuit. The final design is unshielded due to difficulties that the team faced in trying to achieve an MS topology while keeping the peak flux density within the discharge channel. Due to the expected poor performance from this thruster, with the use of Argon. Having a more reliable magnetic field was assessed to be a more appropriate strategy.

The final simulations to confirm the field topology were conducted using COMSOL Multiphysics's 3D magnetic simulations, with applied H-B curves for each material. This allowed for the adjustment of

the channel geometry to account for inaccuracies in the FEMM simulations.

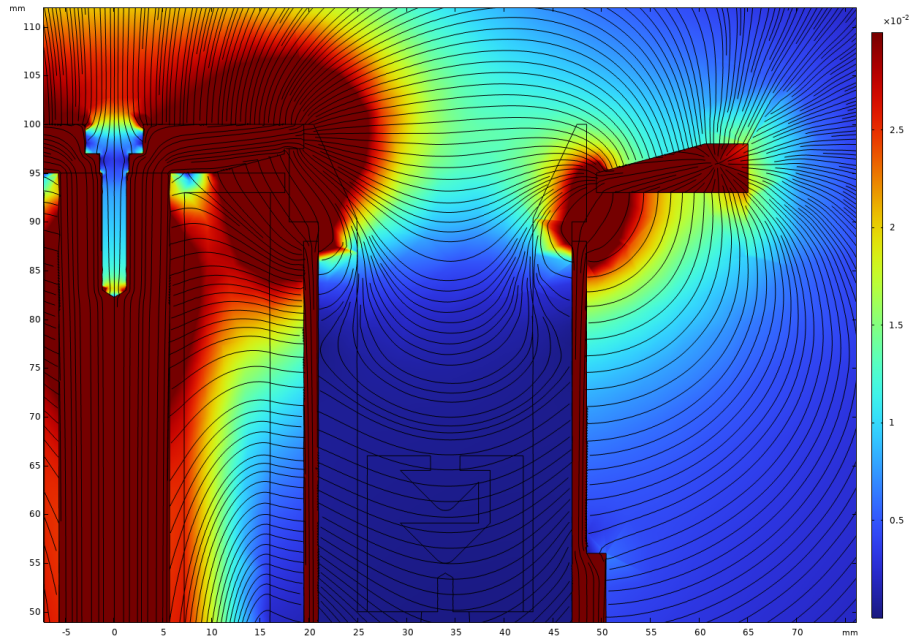


Figure 5: Magnetic Field Contour

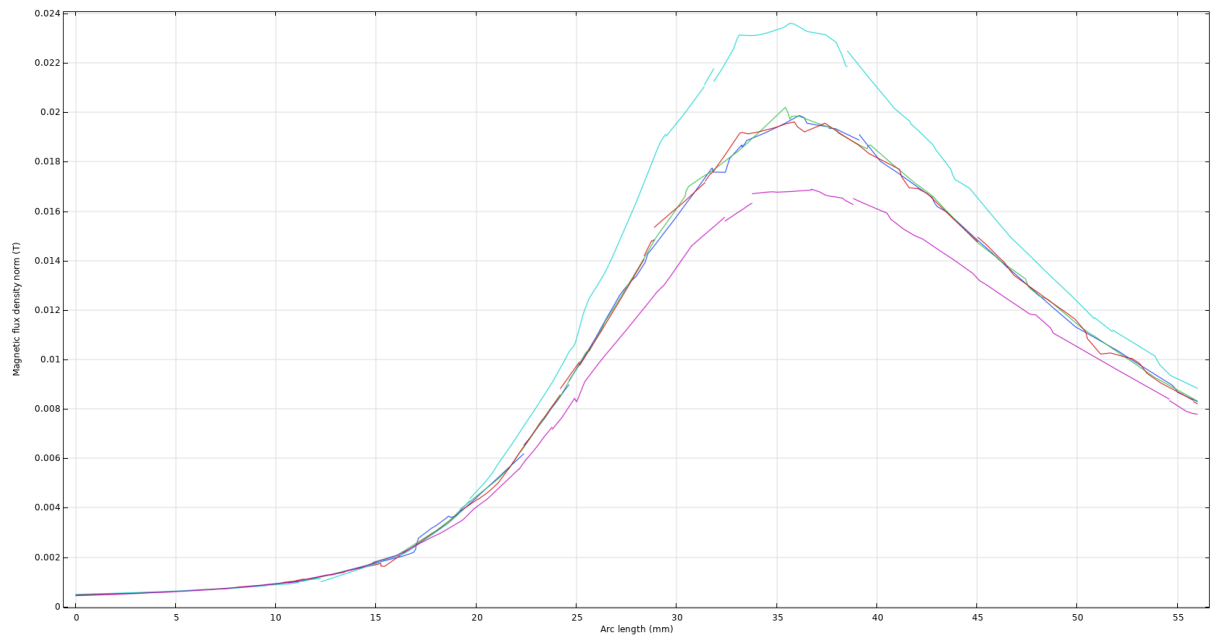


Figure 6: Magnetic Field Strength

The final field topology can be seen in Figure 5, and it is clear from here that the thruster was not MS. Figure 6 displays that the field strength peaked at the channel exit distance. The turquoise line is the field strength along the inside channel wall. The three plots that had an average peak field strength of $0.199T$ were taken at 5° , 10° and 15° along the channel centreline. The purple plot was taken along the outside channel wall. Taking three centerline measurements was necessary to ensure that the magnetic field remained azimuthally identical. This was an initial concern due to using four external electromagnets instead of one external wrap-around. The electromagnets are assumed to all

have a current of 1A. The external electromagnets have 1250 turns each, and the internal electromagnet has 1250 turns. The current in each of the outer electromagnets flows in the same direction, while the current in the central electromagnet flows in the opposite direction. This forces the magnetic field to link across the magnetic circuit.

4 Anode & Gas Distributor Design

This component combines the function of the anode to produce the axial electric field along with that of the gas distributor to equally distribute the gas throughout the discharge channel. Since the flow rate of the gas is very low, it means that the number density of particles will also be low, which will result in few collisions over the discharge channel length. The collisionality of the gas can be described through the Knudsen number, K_n .

$$K_n = \frac{\lambda}{L} \quad (14)$$

In this equation, λ is the mean free path of a neutral atom travelling through the hall thruster, while L is a characteristic length. This could be the discharge channel length, the length of the feed pipe or the internal channels of the gas distributor. Depending on the value of the Knudsen number, different models are required to predict the flow of the gas. COMSOL gives an outline of when different models are appropriate in their Molecular flow user guide, which is summarised in Table 4

Table 4: Types of flow for different Knudsen number ranges [6].

Type of flow	Knudsen Number
Continuum flow	$K_n < 0.01$
Slip flow	$0.01 < K_n < 0.1$
Transitional flow	$0.1 < K_n < 10$
Free molecular flow	$K_n > 10$

To simulate the gas distribution throughout the channel, it was originally assumed that the $k - \omega$ model was appropriate, as in this compact, narrow component, the Knudsen number should be very small to maintain model accuracy. To move into the discharge channel, transitional flow models are likely needed, but as long as the flow uniformly exits the anode, there should be no reason to pursue further simulations downstream.

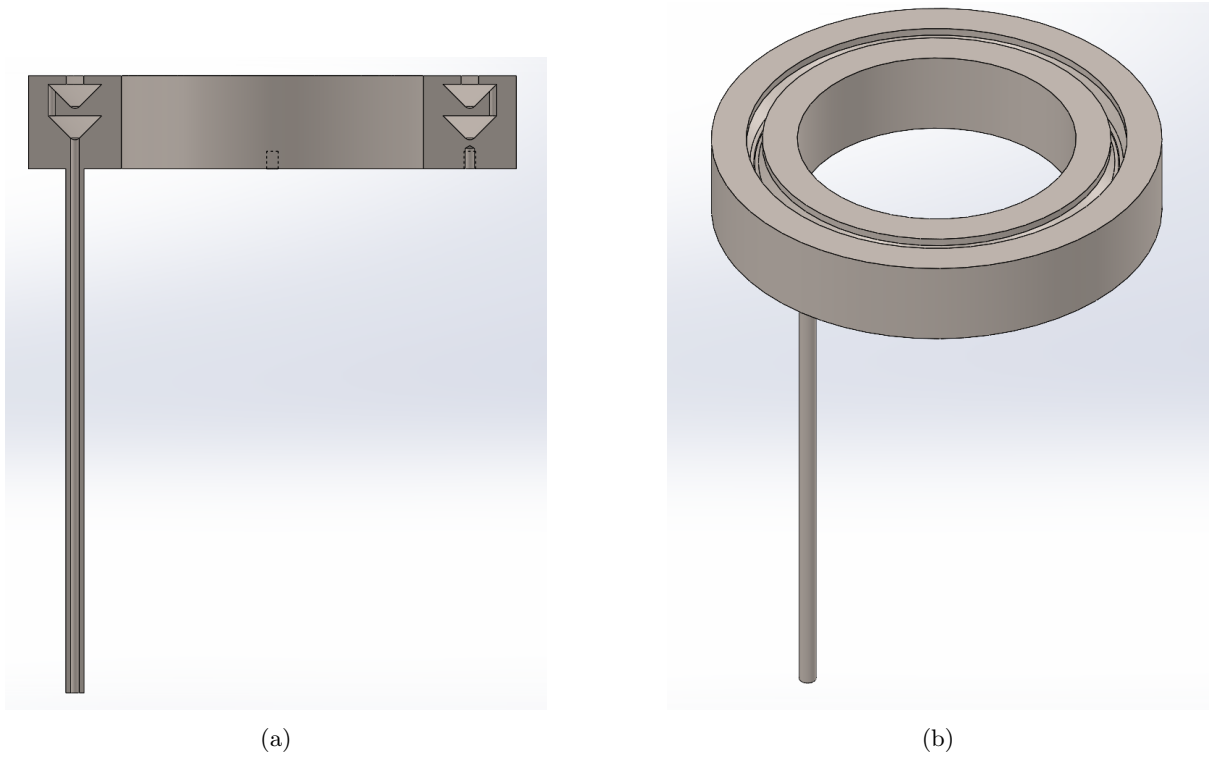


Figure 7: Anode geometry: (a) cross-section view, (b) isometric view.

Figure 7 shows the chosen anode; a 1/8 inch compression fitting will be fitted onto the inlet tube to then allow for 1/4 inch gas supply from the vacuum chamber. Gas will move through the tube and start being distributed by the first tubular ring before moving into the second, where it will hit baffles, further distributing it out before being emitted into the discharge channel. The anode also has 3 mounting holes, which take M2 screws to allow for easy attachment onto the thruster body so that it cannot be removed accidentally. After manufacturing the design, it was realised that the $k - \omega$ model would not be appropriate. The model was re-simulated using the free-molecular-flow model, which would more accurately capture the physics from within the gas distributor.

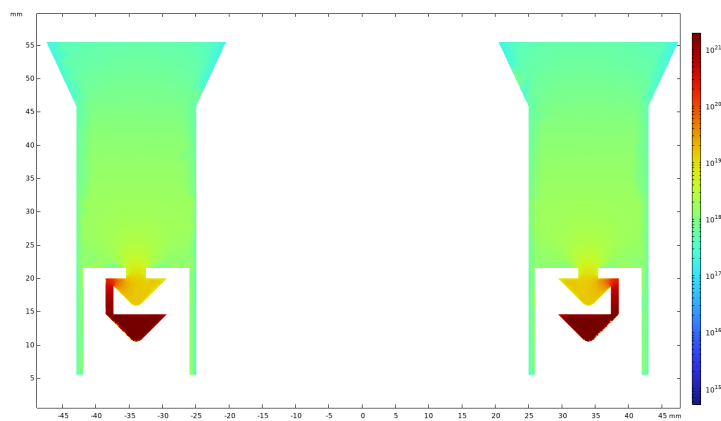


Figure 8: Neutral particle density contour in the discharge channel with a logarithmic colour bar from a cross-sectional perspective.

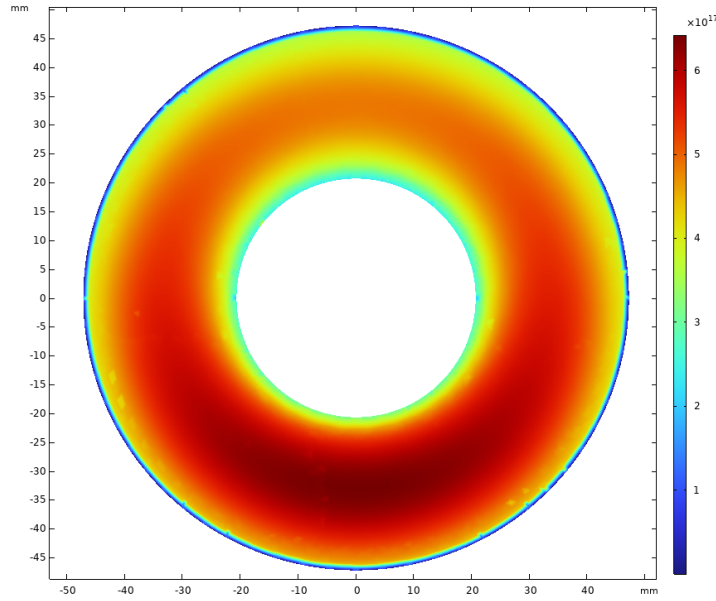


Figure 9: Neutral particle density contour in the discharge channel, at the exit plane.

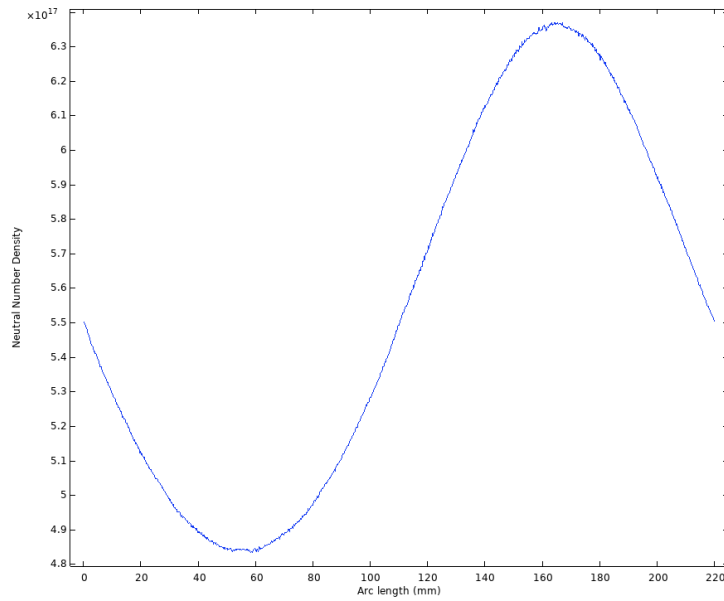


Figure 10: Neutral particle density variation in the discharge channel, at the exit plane.

The data from Figure 8, 9, 10 FMF simulation show that the gas will not be uniformly injected into the discharge channel, with a variation from $(4.85 - 6.35) \times 10^{17}$. This is significant enough to cause an offset thrust vector from the thruster, and so our later designs and models will have more care placed into their anode designs. The issue should not delay our testing of Io-2 as it will still be able to perform and have measurable thrust.

5 Thermal Design

There are several power-loss mechanisms in a Hall-effect thruster; the most notable, which causes significant performance concerns, is thermal heating. To model the thermal loads on the hall thruster channel

walls the following equation is given.

$$P_w = n_e e A_w \left[\left(\frac{kT_e}{e} \right)^{\frac{3}{2}} \left(\frac{2e}{\pi m_e} \right)^{\frac{1}{2}} \exp \left(\frac{e\varphi_S}{kT_e} \right) + \frac{1}{2} \left(\frac{kT_e}{m_i} \right)^{\frac{1}{2}} (\varepsilon - \varphi_S) \right] \quad (15)$$

Where n_e is the number density of electrons, e is the elementary charge, A_w is the surface area of the walls, k is the Boltzman constant, T_e is the electron temperature in Kelvin, m_e is the electron mass, φ_S is the sheath potential relative to the plasma, m_i is the ion mass, ε is the energy of the pre-sheath ion.

Discharge power to the anode can also be calculated and is another mechanism by which heat is transferred to the thruster.

$$P_a \approx 2T_{eV} I_d \quad (16)$$

Where T_{eV} is the temperature in electron-volts and I_d is the discharge current.

A model developed by Marconi C. Porto & et al [7] is used for Io-2 to predict the heat loads experienced by the thruster. The model describes the total wall power losses to have constituents at the outer wall P_o , inner wall P_i , central wall P_c and edge wall P_e .

This relationship is described through the following power balance.

$$P_w = P_o + P_i + P_c + P_e \quad (17)$$

$$n \frac{P_o}{A_o} = \frac{P_i}{A_i} = \frac{P_c}{A_c} = \frac{P_e}{A_e} \quad n = 2 \quad (18)$$

Where A_x is a given area of a wall and x refers to subscripts o, i, c, e with identical meanings as before. As the magnetic field is stronger by the inner wall, it is expected that a larger fraction of the plasma will be concentrated within this region; because of this, the model predicts a power density of the order of 2 : 1 is expected between the inner and outer walls.

Using Equation 15, 16, 17, 18 the discharge heating power is predicted in several regions summarised by Table 5.

Table 5: Table of areas and heating powers used for simulating temperature distribution around the thruster.

Region	Area [mm^2]	Power [W]
Total Wall	15791.16	211.58
Outer Wall	9912.08	96.78
Inner Wall	5445.52	106.34
Center edge	308.21	2.45
Outer edge	125.35	6.02

COMSOL 6.3 was used to thermally simulate the thruster using a solid conduction model with surface-to-surface radiation transfer. The required material parameters for each component are listed

below in Table 6.

Table 6: Material table showing the thermal emissivity (ϵ), thermal conductivity (ω), density (ρ) and specific heat capacity (C_p) of each material used in the simulation

Material	ϵ	ω [W/m ²]	ρ [kg/m ³]	C_p [J/(kg · K)]
1018 mild steel [8], [9]	0.2	49.8	7870	486
copper [8], [10]	0.05	406	8951	377
6061 aluminium [8], [11]	0.1	130	2680	800
HB boron nitride [12]	0.8	20	1900	16100

The resulting simulation gave the following temperature distribution.

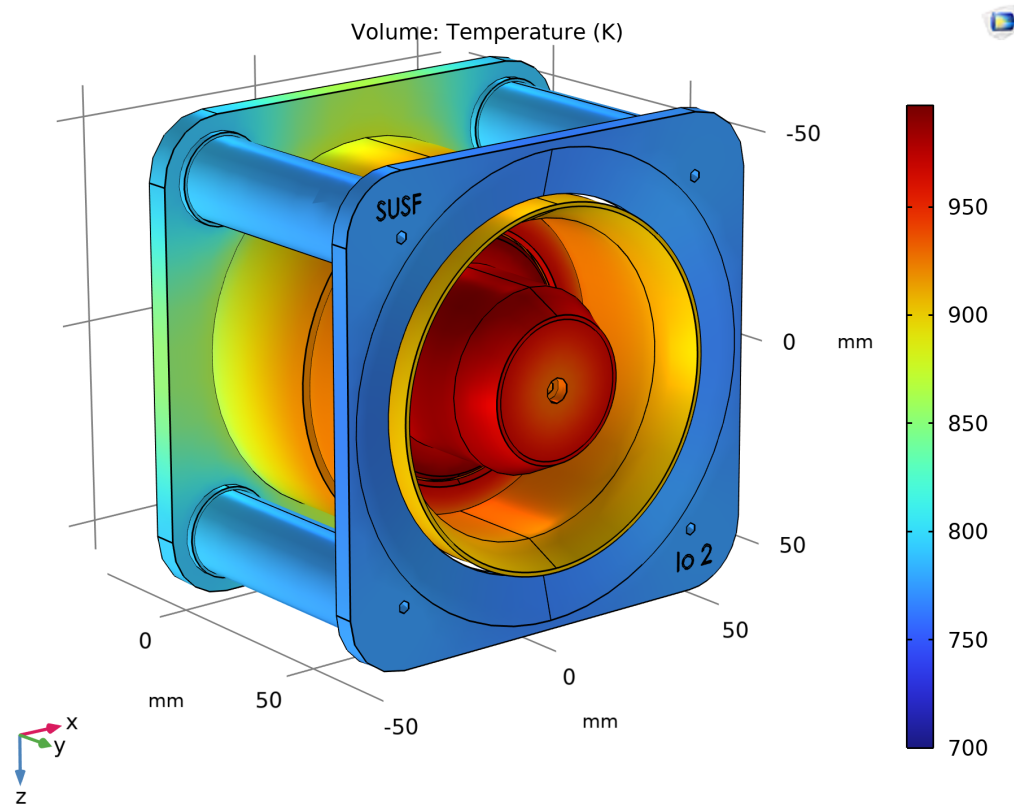


Figure 11: Temperature distribution for Io2.

The model does not account for the thruster stand or thrust balance, so it is likely that the temperatures will be much lower than what is displayed in the simulation. All components remain within their operating temperature limits, illustrating that the thruster should operate safely at the designed 1kW, 300V. To ensure the thruster operates at a lower temperature, the emissivity of all the major steel components will be increased by using low-grit sandpaper to roughen their surfaces.

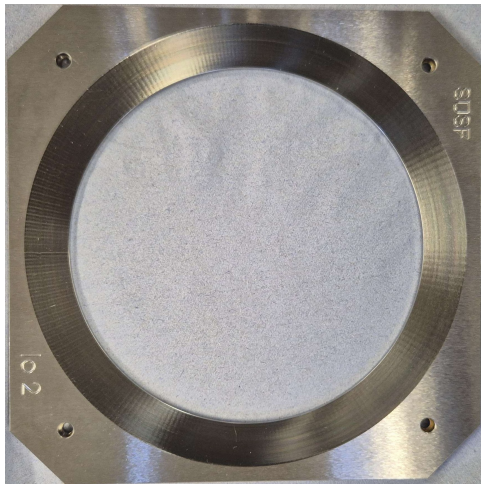
6 Manufacturing

Components for Io-2 were both outsourced and made in-house; some manufacturing capabilities were not available in-house, and specific parts required extremely tight tolerances and complex manufacturing methods. Companies in China were favoured for outsourcing due to previous interactions indicating sufficiently high-quality manufacturing for prices within our budget. Table 7 illustrates the part, manufacturer, and material for each major component of Io-2.

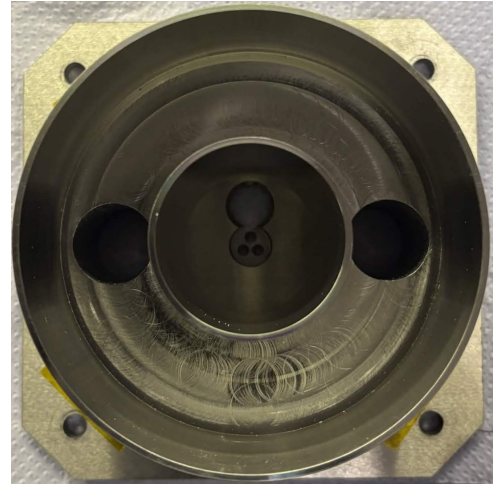
Table 7: Io-2 components manufacturing details

Component	Material	Manufacturer	Cost
Anode gas distributor	316L SS	PCBWay	\$172.25
HB BN Channel	h-BN hexagonal Boron Nitride	Innovacera	\$450.00
Magnetic Screen	1020 Low Carbon Steel	RapidDirect	\$132.64
Top Plate	1020 Low Carbon Steel	RapidDirect	\$61.21
Base Plate	1020 Low Carbon Steel	RapidDirect	\$71.42
Electromagnet Wire	ML Grade 2 Enammelled Copper Wire	wires.co.uk	$3 \times £70 + £117.02$
Thruster stand	6061 Aluminium	RapidDirect	£58.41
Outer bobbins	6061 Aluminium	RapidDirect	$4 \times \$44.89$
Inner bobbin	6061 Aluminium	SUEP	£12.76
Outer Iron cores	1020 Soft Iron	RapidDirect	$4 \times \$36.73$
Inner Iron core	1018 Soft Iron	Craftcloud	£103.02

Once the individual components were acquired or made, a set of cleaning processes were carried out to remove manufacturing grease/oils. All steel components were wiped dry, cleaned with water and IPA and then thoroughly dried again, as shown in Figure 12. To minimise the risk of rusting, the smaller holes ($< \varnothing 3$ mm) were only wiped and cleaned with IPA.



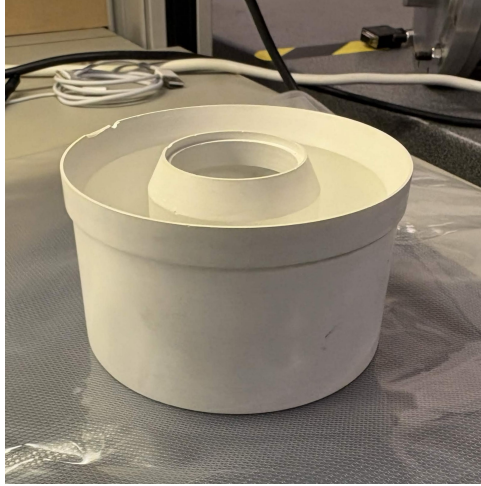
(a) Cleaned top plate



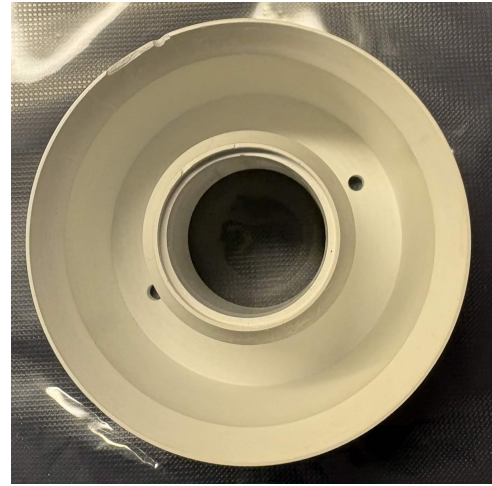
(b) Cleaned thruster body & baseplate

Figure 12: Core metal thruster components post-cleaning

The quality of the Boron Nitride (BN) channel was visually inspected, and it was wiped clean with blue roll. The ceramic channel was unfortunately chipped at the channel exit, as seen in Figure 13. The origin of the chip is a result of a loose metal part being kept with the channel during storage. It is not expected to have a significant impact on the short-term operation of the thruster.



(a) Ceramic Channel



(b) Ceramic Channel

Figure 13: Ceramic Channel

All the electromagnets were wound with 0.56 mm ML Grade 2 enamelled copper wire in-house, on a Douglas ‘automatic’ (hand-cranked) coil winder (shown in Figure 14). The four external electromagnets were wound to 1250 turns, while the central electromagnet had 2000 turns wound. The aluminium bobbins (coil housings) were wrapped with Kapton tape along the inner faces, to help prevent the enamel layer from being scraped off the wire on the bobbin wall and to further electrically insulate the bobbin from the wire. The electromagnets were individually checked for continuity and correct insulation. They were not externally wrapped in Kapton tape due to the insulation provided by ML grade 2 enamel and the positions of the coils being sufficiently far from any steel surfaces; outgassing was also a concern, but the actual risk is undetermined.

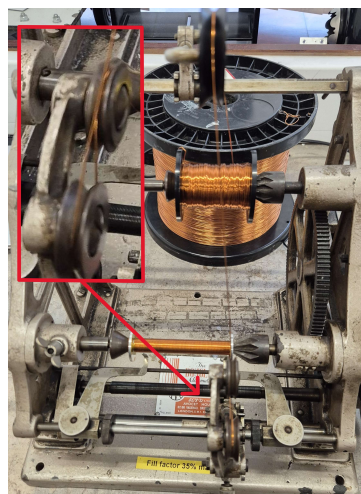
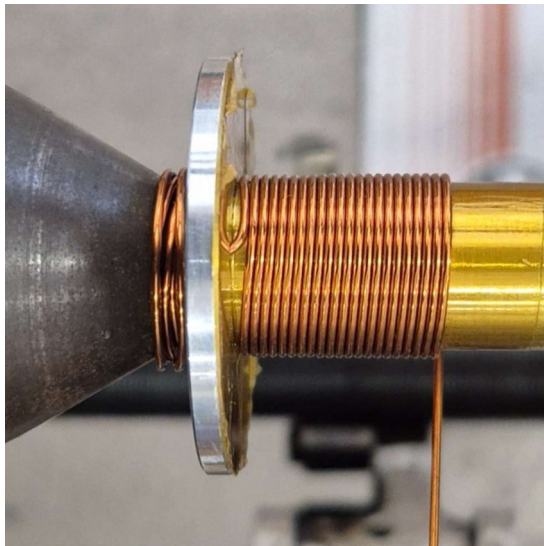
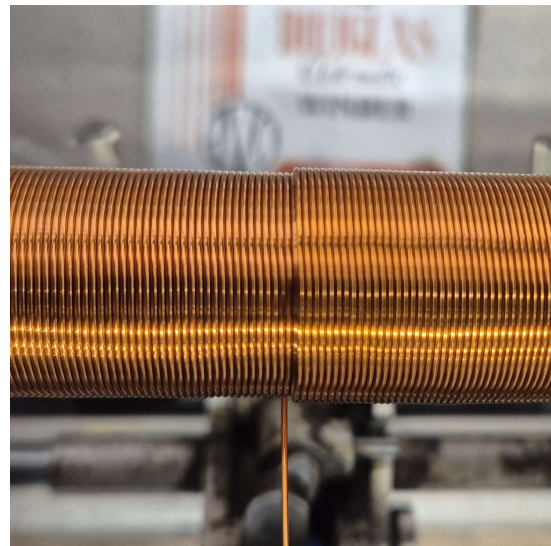


Figure 14: Douglas automatic coil winder

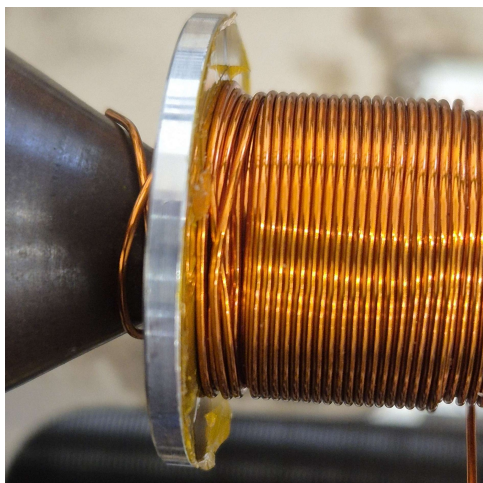


(a) Coil slant error origin

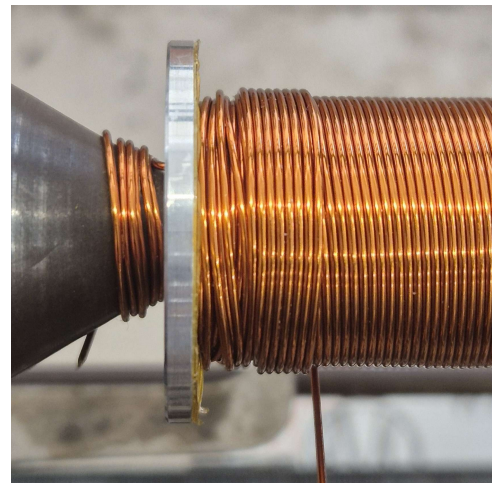


(b) Non-slanted coil section

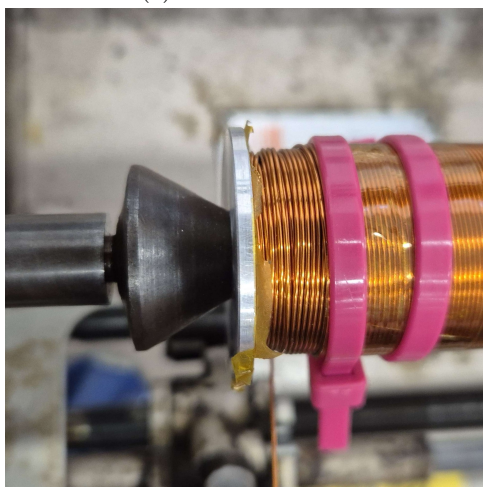
Figure 15: Coil winding setup



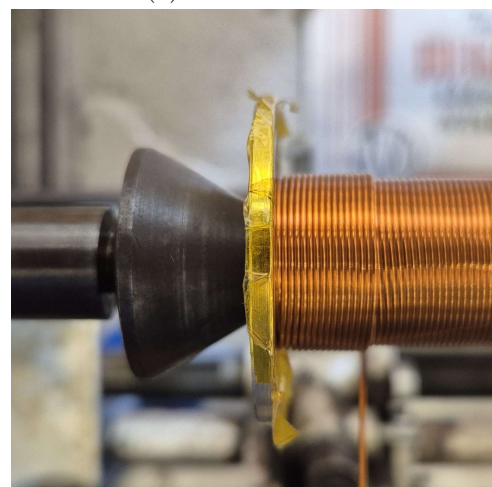
(a) Coil end error 1



(b) Coil end error 2



(c) Coil end error 3 and fix solution

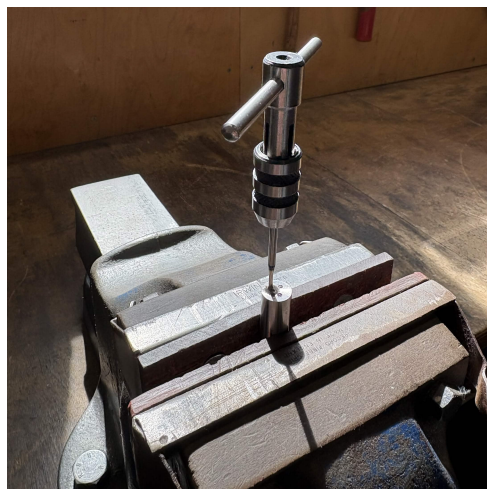


(d) Exemplar coil end

Figure 16: Coil manufacturing errors

Figures 15 and 16, illustrate the major difficulties of coil winding. A natural slant of the coil is present in the coil layers because the origin wire is not at a 90° bend. The wire was not bent this way to prevent the wire from becoming brittle (from work hardening) and exposing the inner copper wire. The slant meant there would be a large gap for the turns on the subsequent layers to slip or sag into. To fix the slant error, the first few layers of each coil were adjusted by hand to force alignment, resulting in non-slanted winds, shown in Figures 15b and 16d.

However, layers of the wire inevitably slipped into gaps due to the weight of the layers above and the slipping of wires that had been manually aligned, not yet falling into the troughs of the layers below. Crossovers occurred as a result of this slipping, as seen in Figures 16a and 16b. The laboratory technician advised that the coils not be unwound to fix these errors, as that could lead to further complications. It was concluded that as long as the number of turns in an area (current density) was maintained as required, these complications would not significantly change the field topology. As seen in Figure 16c, it was later discovered that complications could be fixed or areas re-attempted by holding the accurately wound wire in place with zip-ties and Kapton tape; the coil could be slowly unwound and then re-wound with the aid of tweezers to obtain a better result.



(a) Tapping of central iron core baseplate-connecting holes



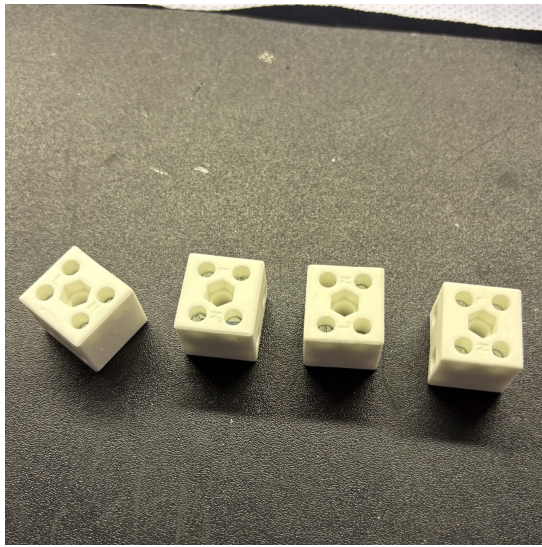
(b) Sheared M2 tap in the inner core

Figure 17: Electromagnet manufacturing

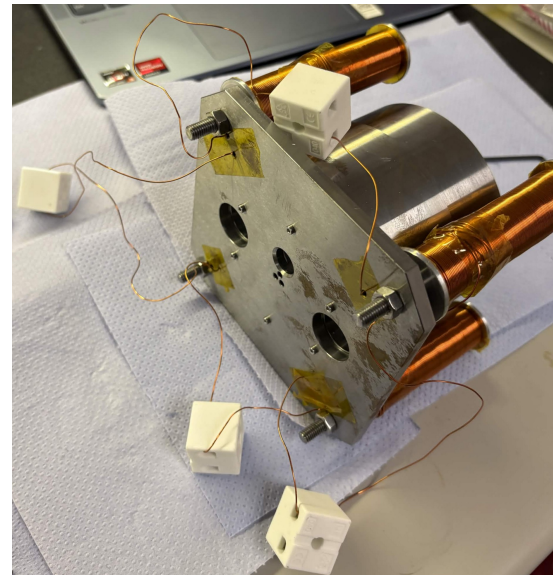
Design changes to the central electromagnet meant the inner iron core was re-fabricated. Unfortunately, the manufacturer did not tap the new inner iron core, and so three M2 holes and one M3 hole was tapped in-house, which is shown in Figure 17a. Even with due care and cutting lubricant, on the final M2 hole, the tap sheared off (shown in Figure 17b). However, this was not a great issue, as many components were designed for errors and failure, and so the two M2 holes will suffice and not interfere with the functionality of the thruster.

Before assembly, any hole through which the wires would pass was covered in Kapton tape to prevent damage to the enamel and to insulate the wire path from the steel components. The electrical circuit utilises ceramic terminal blocks to connect the electromagnets in series, which is demonstrated in Figure 18. To do this, it required the enamel at the connections to be removed. By coating the ends of the wires

in solder, this removes the enamel and provides a suitable contact point for the screws in the terminal blocks.



(a) Ceramic terminal blocks



(b) Electric circuit

Figure 18: Electrical circuit

To prevent surface reactions and to prevent particulates from entering Io-2, the fully assembled thruster was sealed in an airtight partial-vacuum bag and kept in a foam-lined box, as can be seen in Figure 19. Unfortunately, the bag was later found to be pierced, and hence the thruster was exposed to air, causing rust to form on the surface of the metal components. Additionally, one of the wires on the outer electromagnet broke apart. Therefore, a newly wound coil had to be constructed.

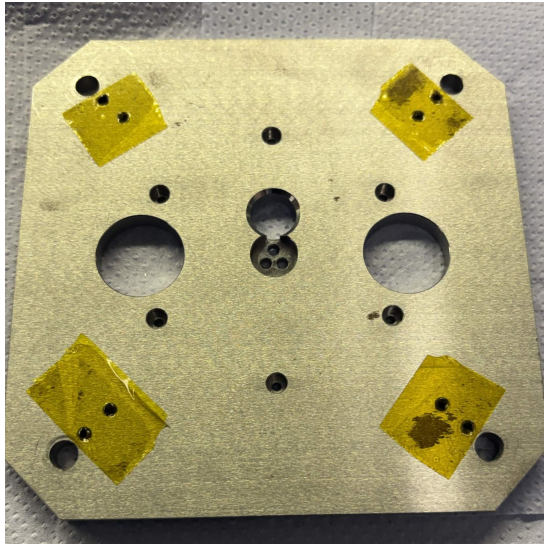


(a) Io-2 Sealed in a vacuum bag

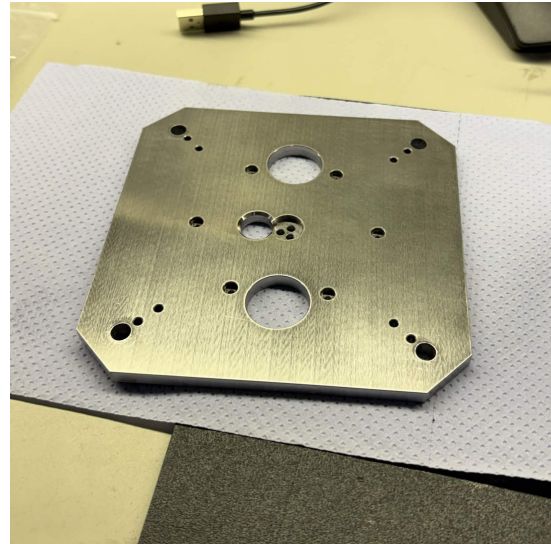


(b) Io-2 Sealed and packed for storage

Figure 19: Storage of Io-2



(a) Rust due to torn vacuum bag



(b) Part cleaned of rust

Figure 20: Baseplate before and after rust removal

The rust on the metal components was removed with 1200 fine-grit sanding sheets (see Figures 20a, 20b) and affected components were re-cleaned with IPA and dried accordingly. There was no rust present on the inside of the metal magnetic screen or anode gas distributor. The broken wire (seen in Figure 21) is believed to be caused by the free rotation and slight movement of the aluminium bobbin. This would result in the wire bending and twisting, and eventually becoming more brittle until failure. This movement perhaps would not have occurred if the sealed bag had not been punctured and an adequate washer fitted to the coil spacers. Ultimately, the broken wire was at the origin of the coil, and hence the coil was required to be re-wound.

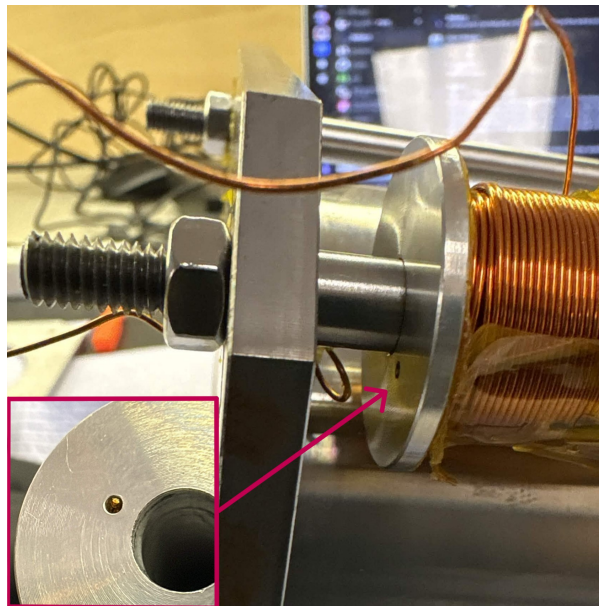


Figure 21: Outer coil clipped wire

To ensure Io-2 maintains suitable thermals, as simulated, the surface was roughened using ... grit

sanding sheets. This roughening process was completed on all the outward-facing surfaces, which includes the front plate, back plate and the outer surfaces of the magnetic screen.

Figure 22 shows the final assembled thruster after the manufacturing stage and before it underwent any tests. This includes the magnetic field test.

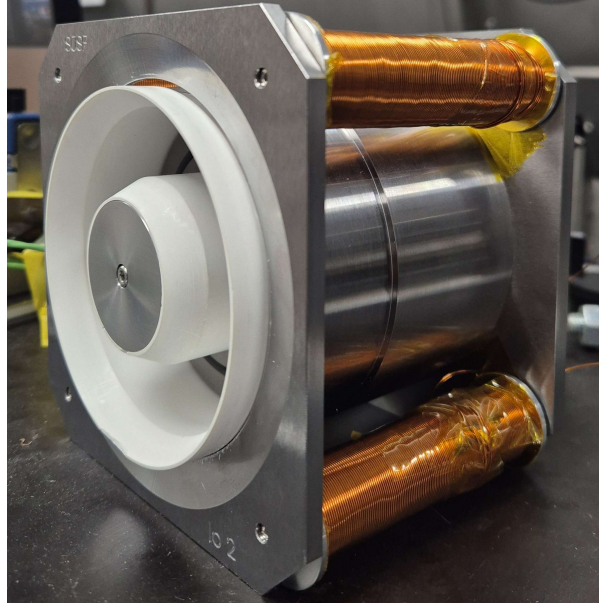


Figure 22: Final Assembled Io-2, pre testing

Listed below are key takeaways from the manufacturing process and resulting changes that could be made to the design and manufacturing stages:

1. Foam inserts or fillings should be used around the thruster before placing it in the sealed vacuum bag, preventing damage to the thruster or the bag.
2. Where possible, bolt holes larger than M2 should be used for ease of tapping and manufacturing.
3. Locking the bobbins' motion and rotation is ideal, preventing the wire from breaking. This can simply be done using additional screws, tapes/adhesives, or even design features such as notches.
4. More careful consideration and research must be conducted on the fabrication method(s) and design to avoid remaking parts.
5. Future design, should try to minimise the number of coil turns, thus reducing manufacturing time and errors.
6. A single outer wrap-around coil could be an alternative to the five external electromagnets, further reducing the chances of errors and manufacturing time.

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